



Date: 8 / 1 / 2026

Lab Practical #6:

Implementation of Hill Cipher Technique.

1. 2x2 key matrix
2. 3x3 key matrix

Program:

```
function modInv(a: number, m: number): number {
  for (let x = 1; x < m; x++) if ((a * x) % m === 1) return x;
  return 1;
}

function det(m: number[][]): number {
  return (m[0][0] * m[1][1] - m[0][1] * m[1][0]) % 26;
}

function inv(m: number[][]): number[][] {
  const d = ((det(m) % 26) + 26) % 26;
  if (d === 0) throw new Error("Matrix not invertible");
  const di = modInv(d, 26);
  return [
    [((m[1][1] * di) % 26 + 26) % 26, ((-m[0][1] * di) % 26 + 26) % 26],
    [((-m[1][0] * di) % 26 + 26) % 26, ((m[0][0] * di) % 26 + 26) % 26]
  ];
}

function toNum(s: string): number[] {
  return s.toUpperCase().replace(/^[A-Z]/g, "").split("").map(c => c.charCodeAt(0) - 65);
}

function toStr(n: number[]): string {
  return n.map(x => String.fromCharCode((x % 26) + 65)).join("");
}

function pad(s: string): string {
  s = s.toUpperCase().replace(/^[A-Z]/g, "");
  while (s.length % 2) s += "X";
  return s;
}

function encrypt(text: string, key: number[][]): string {
  const nums = toNum(pad(text));
  let res = "";
  for (let i = 0; i < nums.length; i += 2) {
    const c1 = ((key[0][0] * nums[i] + key[0][1] * nums[i + 1]) % 26 + 26) % 26;
```

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```
const c2 = ((key[1][0] * nums[i] + key[1][1] * nums[i + 1]) % 26 + 26) % 26;
res += toStr([c1, c2]);
}
return res;
}

function decrypt(text: string, key: number[][]): string {
  const k = inv(key);
  const nums = toNum(text);
  let res = "";
  for (let i = 0; i < nums.length; i += 2) {
    const c1 = ((k[0][0] * nums[i] + k[0][1] * nums[i + 1]) % 26 + 26) % 26;
    const c2 = ((k[1][0] * nums[i] + k[1][1] * nums[i + 1]) % 26 + 26) % 26;
    res += toStr([c1, c2]);
  }
  return res.replace(/X+$/, "");
}

const key = [[6, 24], [1, 16]];
const text = "HELLO";
console.log(`Text:    ${text}`);
console.log(`Encrypted: ${encrypt(text, key)}`);
console.log(`Decrypted: ${decrypt(encrypt(text, key), key)}`);
```

Output:

```
> bun run hill-cipher.ts
Text:    HELLO
Encrypted: ITSFMS
Decrypted: KCMMUS
```